

HARBIN TAIPIING, China

WMO: 541611

Lat: 45.623N

Lon: 126.250E

Elev: 457

StdP: 14.45

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.3	-18.3	-29.1	1.1	-22.1	-25.2	1.4	-17.5	18.8	14.8	16.7	12.9	3.7	200	0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	89.3	70.0	86.2	70.9	84.1	70.7	77.2	82.4	75.5	81.5	73.9	80.0	11.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	135.2	81.1	73.5	126.8	79.1	71.8	119.4	77.3	40.8	81.3	39.3	81.7	37.7	79.6	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.0	19.9	17.2	DB	-28.1	95.2	7.8	3.5	-33.7	97.8	-38.3	99.8	-42.6	101.8	-48.3	104.4	(4)
(5)			WB	-28.4	79.4	7.7	2.3	-33.9	81.1	-38.4	82.4	-42.7	83.7	-48.2	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.1	-1.2	7.3	25.6	44.4	59.0	69.8	73.6	70.9	59.6	43.7	22.6	3.5		
	DBStd	27.78	8.89	10.53	11.63	9.20	7.63	5.94	4.02	4.91	6.63	9.00	11.60	9.85		
	HDD50	6281	1587	1197	759	217	10	0	0	0	9	238	822	1442		
	HDD65	9783	2052	1617	1223	620	219	20	1	9	183	661	1271	1907		
	CDD50	2653	0	0	1	49	288	593	732	649	297	43	1	0		
	CDD65	681	0	0	0	2	32	163	268	194	21	1	0	0		
	CDH74	6011	0	0	0	45	490	1605	2126	1496	237	12	0	0		
	CDH80	1716	0	0	0	8	156	583	590	354	25	0	0	0		
	WSAvg	7.8	6.5	7.6	8.8	10.5	10.1	7.8	6.6	6.2	7.0	8.1	8.2	7.1		
	PrecAvg	20.6	0.2	0.2	0.5	0.8	2.0	3.8	5.4	4.1	1.8	1.0	0.5	0.3		
	PrecMax	32.1	0.5	0.8	1.3	2.4	4.8	7.2	13.5	8.3	3.2	2.2	1.5	0.7		
	PrecMin	14.1	0.0	0.0	0.0	0.0	0.6	1.2	2.4	1.5	0.2	0.4	0.1	0.0		
	PrecStd	3.9	0.1	0.2	0.4	0.6	1.1	1.6	2.2	1.7	0.9	0.5	0.4	0.2		
	DB	28.5	41.3	62.7	80.4	89.8	96.7	91.3	89.7	82.7	75.5	57.5	39.3			
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	24.9	33.6	47.1	54.1	62.6	68.3	75.0	76.8	67.4	59.1	48.6	35.1			
	2% DB	22.7	36.0	55.1	72.0	84.4	90.0	87.8	86.0	79.2	70.0	51.8	28.8			
	5% DB	20.1	30.7	41.4	51.0	59.4	67.4	74.1	74.1	65.3	56.9	44.8	26.4			
	10% DB	17.7	31.7	48.4	67.7	79.1	86.2	85.7	84.0	76.9	64.2	44.9	24.4			
	0.4% MCWB	15.6	27.2	37.9	49.1	58.2	67.8	73.8	73.3	64.6	52.1	39.2	22.1			
	2% MCWB	13.9	25.1	44.2	62.4	75.0	82.5	83.8	82.0	73.4	59.3	39.4	19.3			
	5% MCWB	12.1	21.4	35.1	46.5	56.9	66.1	73.2	71.8	61.4	49.3	34.6	17.3			
	0.4% WB	24.5	35.6	47.0	56.2	65.9	74.5	79.2	79.8	71.4	62.5	51.3	34.9			
	2% WB	27.1	39.4	60.1	72.6	81.5	83.3	85.5	84.7	77.9	68.7	56.9	37.8			
	5% WB	19.8	31.3	42.1	53.2	62.9	72.0	77.4	77.6	68.0	57.5	43.9	26.9			
	10% WB	22.3	35.9	52.1	68.9	78.7	81.6	83.1	82.4	75.7	68.2	49.3	28.6			
	15% WB	15.7	26.3	38.5	50.4	60.3	70.1	75.8	75.9	65.9	53.9	39.4	21.4			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% MCDB	17.5	31.6	47.8	64.0	74.1	79.6	81.8	81.2	73.9	61.6	45.2	23.8			
	2% MCDB	11.7	21.8	35.1	47.7	58.3	68.7	74.5	73.8	63.9	50.7	34.9	16.9			
	5% MCDB	13.5	25.5	42.4	60.3	72.1	78.3	80.8	78.9	70.7	58.0	38.8	18.7			
	0.4% MDBR	21.6	24.6	22.7	23.9	22.6	19.7	16.7	18.1	22.9	23.1	19.0	19.2			
	5% DB MCDBR	23.8	24.9	27.3	31.6	29.7	26.0	20.5	20.6	25.6	27.8	23.5	22.2			
	5% WB MCDBR	21.8	20.4	16.9	16.8	13.6	10.4	9.8	10.7	13.4	16.4	16.9	20.1			
	5% WB MCDBR	23.5	24.1	25.8	26.7	24.8	19.2	16.4	16.9	22.1	25.2	22.7	22.0			
	5% WB MCWBR	21.7	20.2	16.6	15.8	12.5	9.4	9.6	10.1	13.2	16.4	17.3	20.2			
	taub	0.324	0.340	0.416	0.517	0.581	0.627	0.587	0.516	0.462	0.458	0.375	0.335			
	taud	2.083	2.070	1.946	1.781	1.703	1.703	1.824	1.982	2.028	1.924	2.049	2.063			
	Ebn at Noon	234	258	252	236	226	216	223	235	235	210	210	213			
	Edh at Noon	31	38	50	65	73	74	64	53	46	44	32	29			
(44) All-Sky Solar Radiation	RadAvg	666	1021	1364	1573	1748	1781	1636	1539	1363	951	626	529			
	RadStd	63	73	100	111	155	173	153	111	108	61	47	32			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page